

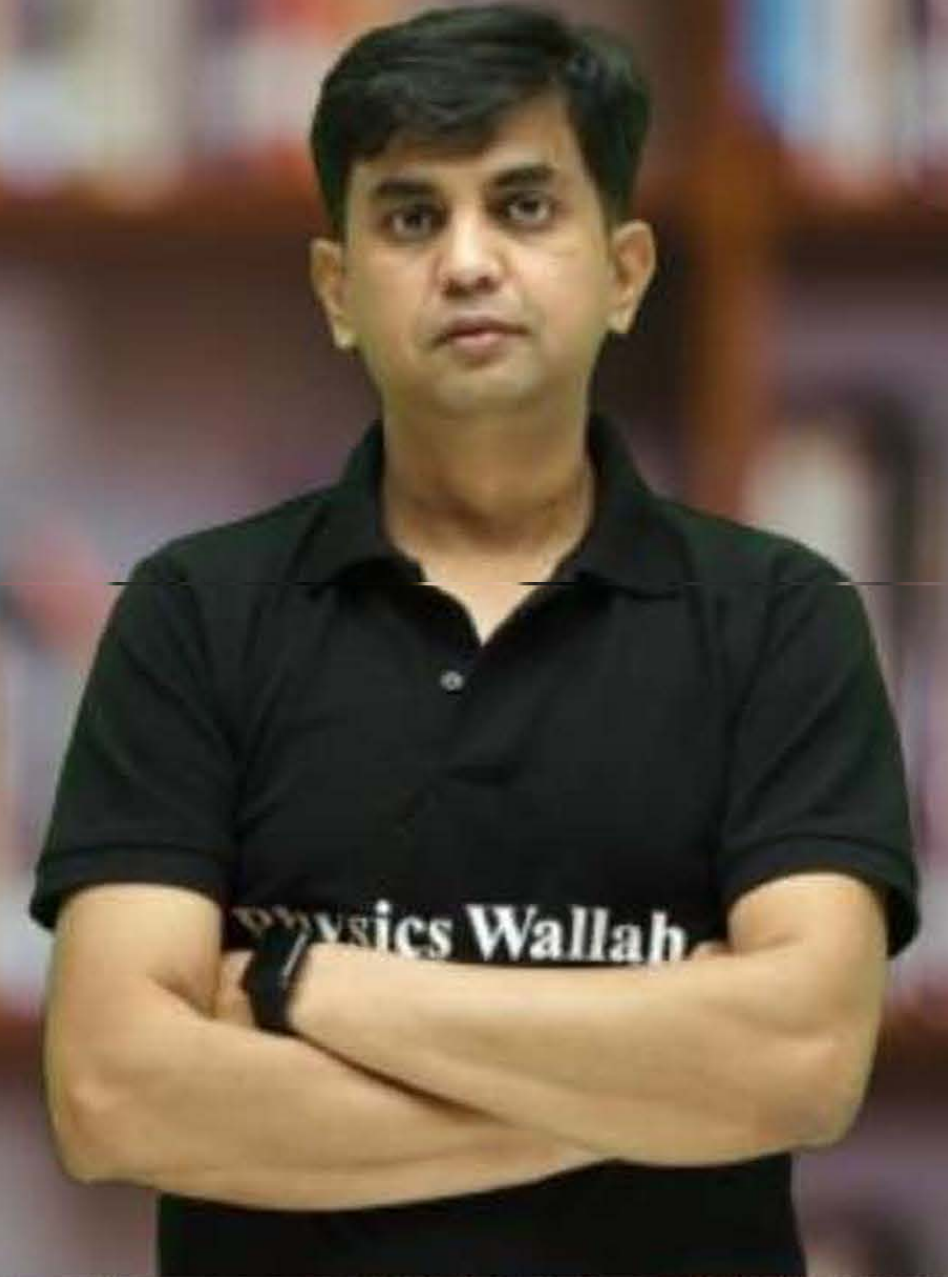
CS & IT ENGINEERING



Computer Network-1

Routing

DPP - 01



By- Abhishek Sir

[MCQ]



#Q. Link state routing is based on _____.

☒ **A**

Dijkstra's algorithm

☐ **B**

Bellman-Ford algorithm

☐ **C**

Floyd-Warshall algorithm

☐ **D**

Flooding

Ans: A

[MCQ]



#Q. Distance vector routing is based on _____.

~~A~~

Dijkstra's algorithm

B

Bellman-Ford algorithm

~~C~~

Floyd-Warshall algorithm

~~D~~

Flooding

Ans: B

[MCQ]



#Q. _____ is a link state routing protocol.

A RIP (Routing Information Protocol) $\Rightarrow$ DV Routing

✓ **B** OSPF (Open Shortest Path First) $\Rightarrow$ LS Routing

C BGP (Border Gateway Protocol)

D IS-IS (Intermediate System to Intermediate System)

Ans: B

[MCQ]



#Q. _____ is a distance vector routing protocol.

☒ **A**

RIP (Routing Information Protocol)

☐ **B**

OSPF (Open Shortest Path First)

☐ **C**

BGP (Border Gateway Protocol)

☐ **D**

IS-IS (Intermediate System to Intermediate System)

Ans: A

[MCQ]

#Q. 'Count to infinity' problem is associated with which routing algorithms ?

☒ **A** Distance Vector Routing

☐ **B** Link State Routing

☐ **C** Path Vector Routing

☐ **D** None of the above

Ans: A

[MCQ]

#Q. Identify correct statement regarding 'Link state' routing algorithm.

- A** Dijkstra Algorithm executed by any one router FALSE
- B** Dijkstra Algorithm executed by a particular routers FALSE
- ☒ **C** Dijkstra Algorithm executed by all the routers TRUE
- D** No need to run Dijkstra Algorithm explicitly. FALSE

Ans: C

[MCQ]



#Q. Consider a network of three routers A, B, and C, where B is connected to A and C with distance 4 and 5 respectively.

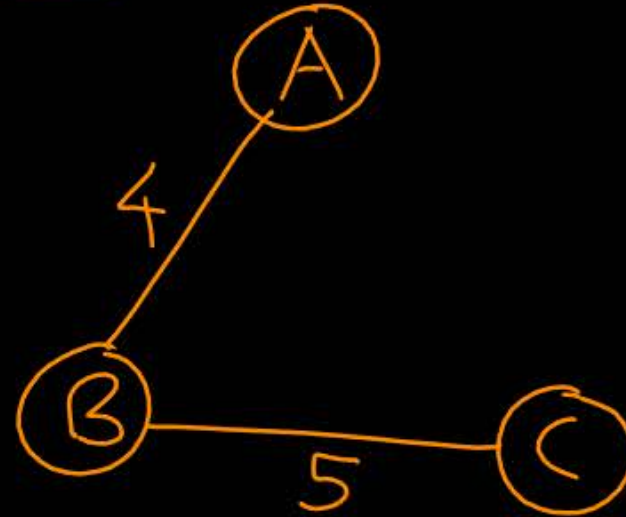
Identify correct distance vector of router B after one iteration (round of update) of the distance vector routing algorithm?

~~A~~ $A = 5, B = 0, C = 4$

~~B~~ $A = 4, B = 0, C = \infty$

~~C~~ $A = \infty, B = 0, C = 5$

~~D~~ $A = 4, B = 0, C = 5$



$$D.V.(B) = [D_B(A) = 4, D_B(C) = 5]$$

Ans: D

[MCQ]



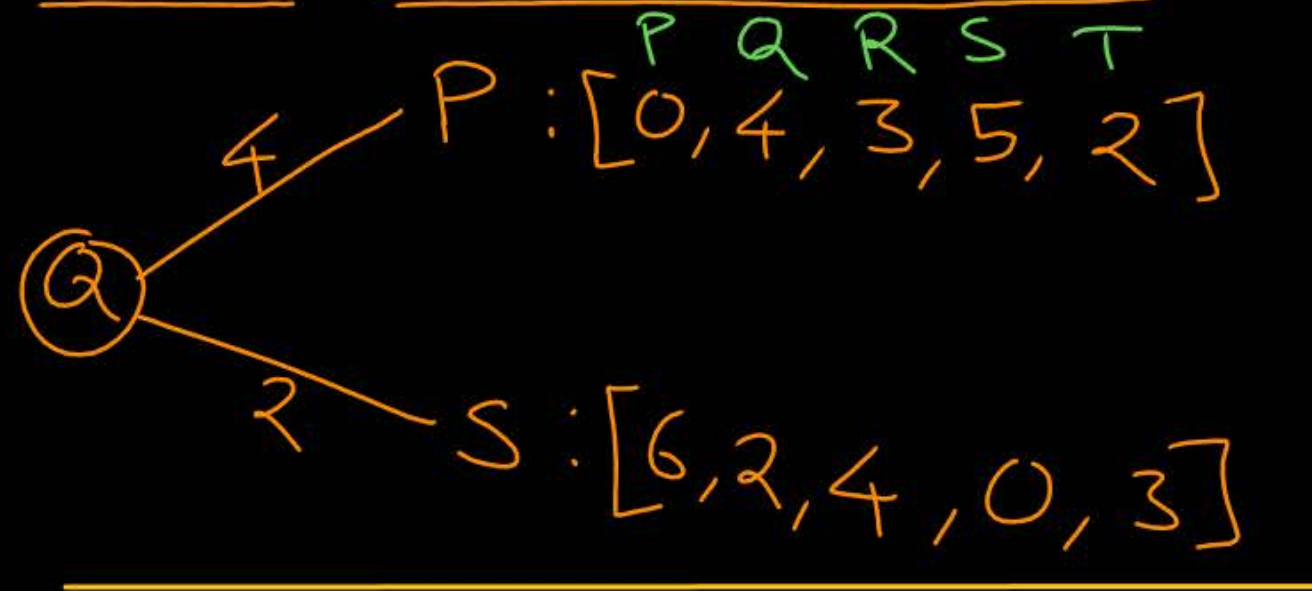
#Q. Consider a network of five routers : $[P, Q, R, S \text{ and } T]$ uses 'Distance Vector' routing algorithm. Every router maintain distance to other router in the above mentioned routers sequence. Router Q is directly connected with router P and S with distance 4 and 2 respectively. If distance vector of P : $[0, 4, 3, 5, 2]$ and distance vector of S : $[6, 2, 4, 0, 3]$ are arrived at router Q, then what should be router Q updated distance vector?

~~A~~ $[4, 0, 6, 2, 6]$

~~B~~ $[4, 0, 7, 2, 5]$

~~C~~ $[4, 0, 6, 2, 5]$

~~D~~ $[4, 0, 7, 2, 6]$



P Q R S T
Q: $[4, 0, 6, 2, 5]$
(P) (Q) (S) (S) (S)

Ans: C

[MCQ]



#Q. Consider a network of five routers : [A, B, C, D and E] uses 'Distance Vector' routing algorithm. Every router maintain distance to other router in the above mentioned routers sequence. Router C is directly connected with router A and B with distance 3 and 5 respectively. If distance vector of A : [0, 5, 3, 4, 2] and distance vector of B : [6, 0, 4, 5, 3] are arrived at router C, then what should be router C updated distance vector?

~~A~~

[0, 3, 5, ∞ , ∞]

~~B~~

[3, 5, 0, ∞ , ∞]

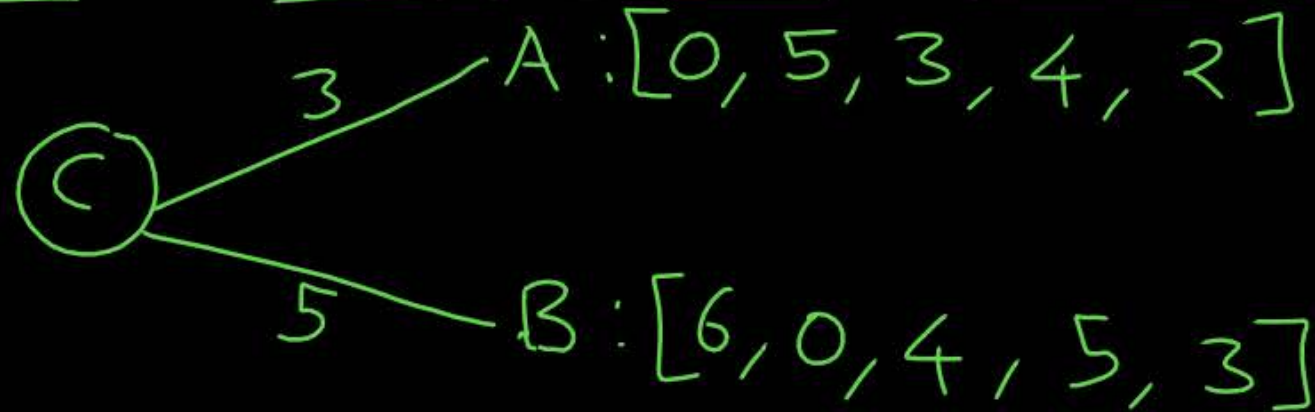
~~C~~

[3, 5, 0, 10, 6]

☒ D

[3, 5, 0, 7, 5]

Ans. D



A B C D E
C : [3, 5, 0, 7, 5]
(A) (B) (C) (A) (A)

[MSQ]



#Q. Which of the following statements is/are correct regarding distance vector and link state routing protocols?

☒ A Distance vector routing protocol is prone to routing loops than link state routing protocol. TRUE

☒ B Link state routing protocols generally consume more bandwidth than distance vector routing protocols. TRUE

☐ C Distance vector routing protocols converge faster than link state routing protocols. FALSE

☒ D Link state routing protocols ensure that each router possesses a consistent and identical view of the network topology. TRUE

Ans: A, B & D



THANK - YOU